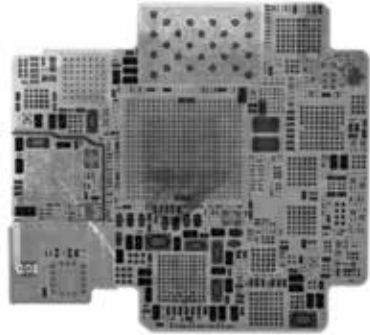


3. **SiP:** System-in-Package is an MCM which is stacked vertically. Traditionally MCMs were horizontal – if there were multiple chips on the same PCB, MCM would mean bringing them on the same substrate layer and connecting them on a single package. But that means wastage of space on the PCB. To counter this, different ICs could be stacked on top of one-another and connected vertically rather than horizontally. Such a vertical MCM is termed SiP. You can think of a SiP as a 3D MCM!
4. **SoP:** In terms of area utilised on a PCB, 90 per cent of the space is occupied by passive components and their interconnections. SoP is the process to scale them down into thin-film components and fit them on one chip. Apple Watch is one fine example of an SoP.
5. **SoM:** System-on-module are complete systems on one module. Here the module is generally a single PCB on which multiple components (which might constitute a SoC) are attached. These units are normally complete systems and are designed for specific purposes. You can think of the Macbook's logic board as a SoM – it contains the processor, graphics unit, CPU, RAM, I/O controllers all on one PCB. You can attach other peripherals to it and it behaves as a complete computer.



Apple Watch – a Mini Computer on your wrist is a SoP – it has everything as part of one single package

If you read those terms carefully, you might have noticed one thing: they are all very similar (except for SoM, which is a full PCB, not a single package). In this regard, the word 'SoC' is more of an umbrella term. An implementation where you can fit in multiple circuits on a single package can be termed as an SoC. However, the way we generally try to define the new in terms of the old and yet want to differentiate among them, the terminology around SoCs has not evolved in a very friendly one.

What hardware do SoCs consist of?

A lot of different modules come together to form an SoC and there can be a lot of variations therein. For example – the lift controller requires one type